

SmartCare v0.4 - Domain Implementation Workbook

Week 7 student resource

1. UML-to-Code Trace

UML element	Python element	Implemented?	Notes
Patient class	class Patient:	Yes	Matches approved domain model.
Patient - id, - name	self.id, self.name	Yes	Initialized in <code>__init__</code> .
Patient + getHistory ()	def get_history(self):	Yes	Returns the appointments list.
Practitioner class	class Practitioner:	Yes	Matches approved domain model.
Practitioner + getSchedule ()	def get_schedule(self):	Yes	Returns the appointments list.
Appointment class	class Appointment:	Yes	Matches approved domain model and includes protected state.
Appointment - status	self.status	Yes	Uses AppointmentStatus Enum.
Association (1-to-many)	self.appointments = []	Yes	Used lists in Patient and Practitioner.

2. Domain Invariants

Class	Invariant / rule	How protected
Patient	Name and contact info cannot be blank	Checked with an if statement in <code>__init__</code> that raises a <code>ValueError</code> .
Appointment	Status cannot change if already cancelled	Checked in the <code>cancel ()</code> method, raises an <code>Exception</code> if already cancelled.

3. Composition / Inheritance Decisions

Relationship	Decision	Rationale
Patient / Appointment	Composition/Association An Appointment is not a type of	Patient; it is a separate event linked to a Patient.
Practitioner / Appointment	Composition/Association	An Appointment is linked to a Practitioner, not inherited from one.

4. AI Pair-Programming Record

AI contribution	Conforms?	Decision	Reason	Verification
Appointment class	Yes	Accepted	Matches UML attributes perfectly.	Visual code inspection.
AppointmentStatus Enum	Yes	Accepted	Safely restricts status to specific options.	Tested valid transitions.
Cancel () method logic	Yes	Accepted	Protects the cancelled state invariant.	Manual behaviour check script.

5. Updated UML

No updates are necessary since the implementation successfully created directly from the approved UML without uncovering any justified design changes.